

# Normalization “bloopers”

1)

## Original table

BEVERAGE [ BevCode, BevName , Category, Price]

## 1NF

BEVERAGE [ BevCode, BevName , Category, Price]

### **Partial Dependencies**

BevCode → BevName,

Category → Price

### **New Tables**

BEVERAGE [ BevCode, BevName]

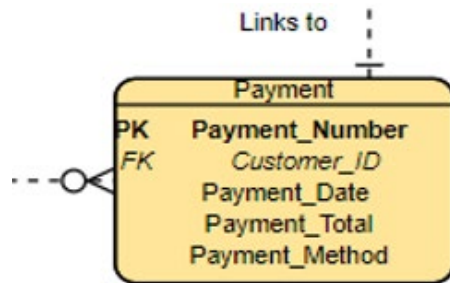
CATEGORY [ Category, Price]

**WHAT** Partial Dependencies??? There is NO concatenated PK!!

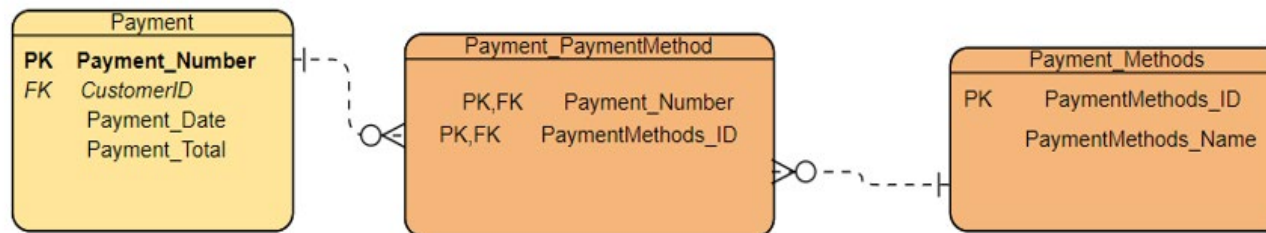
One of the pillars of normalization is that tables are “split” without the loss of information. From these two tables, it is impossible to get the **Category** or **Price** of any **BevCode**.

VERDICT: This BEVERAGE table should have remained as it was.

2)



## Normalization of Payment Entity

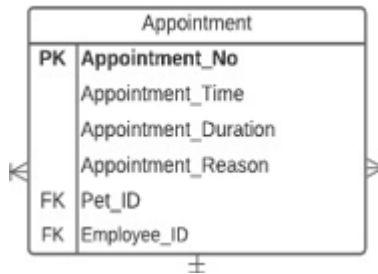


The original table should have contained the attributes: **Payment number**, **Customer\_ID**, **Payment\_date**, **Payment\_Total**, **Payment\_Method\_ID**, **Payment\_Method\_Name**.

Then the TD Payment\_Method\_ID → Payment\_Method\_Name could be removed and one would have 2 resulting tables:

PAYMENT[Payment number, Customer\_ID, Payment\_date, Payment\_Total, Payment\_Method\_ID] and  
 PAYMENT\_METHOD[Payment Method\_ID, Payment\_Method\_Name]

3)



### Entity Normalization

#### 1<sup>st</sup> Normal Form

| Appointment_No | Time  | Duration   | Reason           |
|----------------|-------|------------|------------------|
| 0001           | 09:00 | +/- 30 min | Dental           |
| 0002           | 10:30 | +/-45 min  | Allergies        |
| 0003           | 12:00 | +/-15 min  | General Check-up |
| 0004           | 13:45 | +/-15min   | General Check-up |

#### 2<sup>nd</sup> Normal Form

| Appointment_No | Time  |
|----------------|-------|
| 0001           | 09:00 |
| 0002           | 10:30 |
| 0003           | 12:00 |
| 0004           | 13:45 |

The Duration of the appointment is partially dependent on the Reason for the appointment.

| Reason           | Duration   |
|------------------|------------|
| Dental           | +/- 30 min |
| Allergies        | +/-45 min  |
| General Check-up | +/-15 min  |
| General Check-up | +/-15min   |

#### 3<sup>rd</sup> Normal Form

| Appointment_No | Time  |
|----------------|-------|
| 0001           | 09:00 |
| 0002           | 10:30 |
| 0003           | 12:00 |
| 0004           | 13:45 |

| Reason           | Duration   |
|------------------|------------|
| Dental           | +/- 30 min |
| Allergies        | +/-45 min  |
| General Check-up | +/-15 min  |
| General Check-up | +/-15min   |

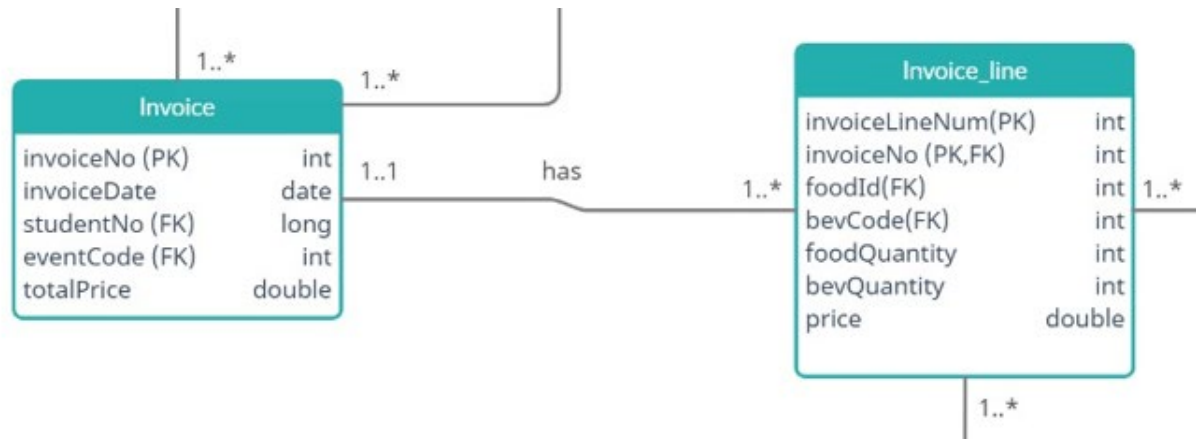
There are no transitive dependencies from the 2<sup>nd</sup> Normal Form.

Notice how the number of attributes reduced from the ERD to the start of Normalization..... What has happened to the missing attributes??? Without **Pet\_ID** and **Employee\_ID**, how are you going to know which pet the appointment was for, or who attended to them???

**Duration** might be dependent on **Reason**, BUT it is NOT a PD, it is a TD!!  
For a PD, the determinant must be PART OF the PRIMARY KEY and you can only have **parts** if you have a **concatenated PK!!!**  
The dependency **Reason** → **Duration** is a Transitive Dependency

The 3NF tables should be:  
**APPOINTMENT**[Appointment\_No, Time, Reason, Pet\_ID, Employee\_ID]  
and  
**APPT\_REASON**[Reason, Duration]

4)



## Normalisation

### Invoice Table:

#### 1NF

INVOICE [InvoiceNo, StudentNo, EventCode, InvoiceDate, TotalPrice]

#### 2NF

INVOICE [InvoiceNo, StudentNo, EventCode, InvoiceDate, TotalPrice]

#### Partial Dependencies

InvoiceNo → InvoiceDate, TotalPrice

Event → EventCode, StudentNo

#### New Tables:

INVOICE [InvoiceNo, InvoiceDate, TotalPrice]

EVENT [EventCode, StudentNo]

- No Transitive Dependencies therefore the tables will not go to 3NF.

**WHAT** Partial Dependencies??? There is NO concatenated PK!!

Where does this **Event** attribute/determinant come from? It's not in the table.

No TDs mean that the table **IS** in 3NF!

VERDICT: The INVOICE table should remain as is, it is in 3NF

## Invoice\_Line Table:

1NF:

INVOICE\_LINE [InvoiceLineNum, InvoiceNum, FoodId, BevCode, FoodQuantity, BevQuantity, Price]

The PK should be **InvoiceNum + InvoiceLineNum**

2NF

INVOICE\_LINE [InvoiceLineNum, InvoiceNum, FoodId, BevCode, FoodQuantity, BevQuantity, Price]

Partial Dependencies:

FoodId → FoodQuantity, Price

BevCode → BevQuantity, Price

New Tables:

INVOICE\_LINE [InvoiceLineNum, InvoiceNum]

FOOD [FoodId, FoodQuantity, Price]

BEVERAGE [BevCode, BevQuantity, Price]

No Transitive Dependencies therefore the tables will not go to 3NF.

**WHAT** Partial Dependencies??? There is NO concatenated PK!!

If these 2 were REAL dependencies, it would mean that EVERYONE that ordered a cheese burger (a certain FoodID) would ALWAYS order the SAME FoodQuantity! And the same for the beverages...

No TDs mean that the table **IS** in 3NF!

**VERDICT:** The table should remain as is, with the **concatenated PK** mentioned above.